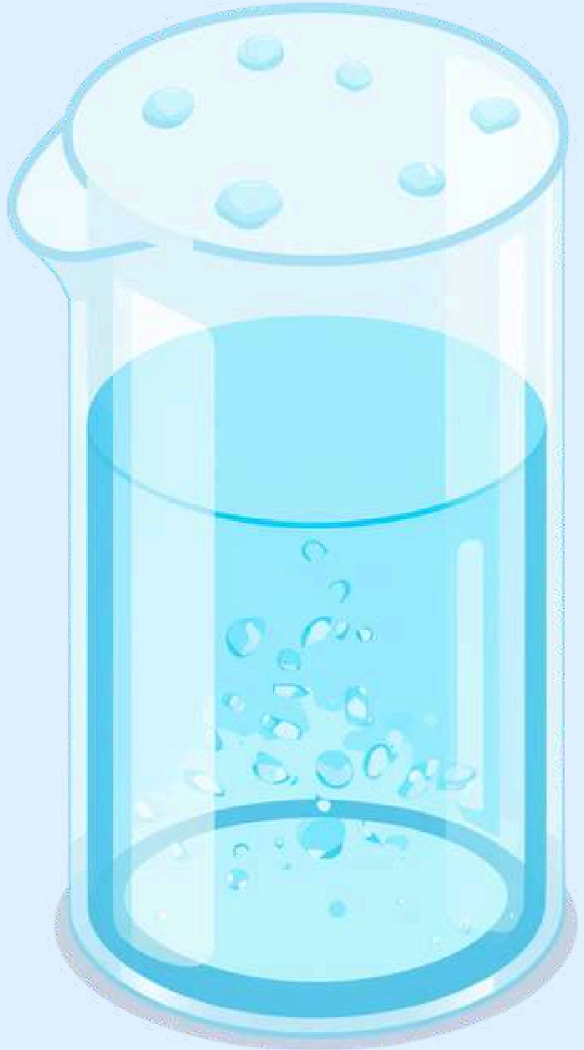
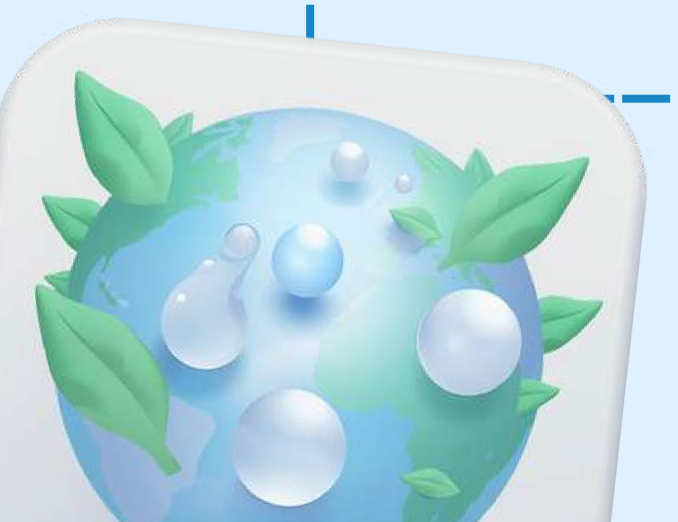
ENVIRONMENTAL CHEMISTRY: WATER QUALITY TESTING



What Is Environmental Chemistry?

Environmental chemistry is the study of chemicals in nature and how they affect our environment. Scientists examine how substances move through air, water, and soil, and how they impact plants, animals, and people. Think of it as being a detective for nature! Environmental chemists investigate what happens when different chemicals enter rivers, forests, and oceans. They help us understand how to keep our planet healthy and safe for all living things.



Why Do We Test Water?

Over 2 billion people worldwide lack access to safe drinking water. Testing water helps us identify harmful contaminants before they cause illness. By checking water quality, we can protect our families, communities, and the environment. Regular testing ensures the water we drink, cook with, and use daily is safe for everyone.

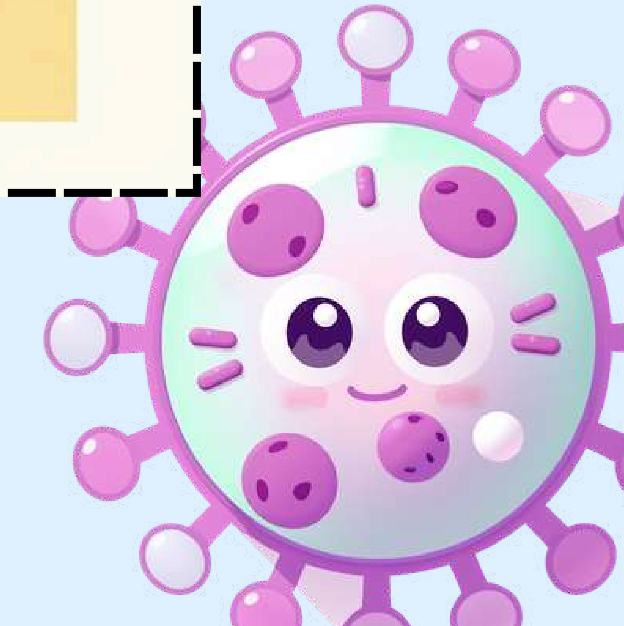
What Is Water Quality?

Clean water is safe for drinking, cooking, and supporting life. Water quality tells us how healthy water is based on three key aspects:

• Physical: How water looks, smells, and tastes
• Chemical: Substances dissolved in water like minerals
• Biological: Living things like bacteria in water

Common Water Contaminants

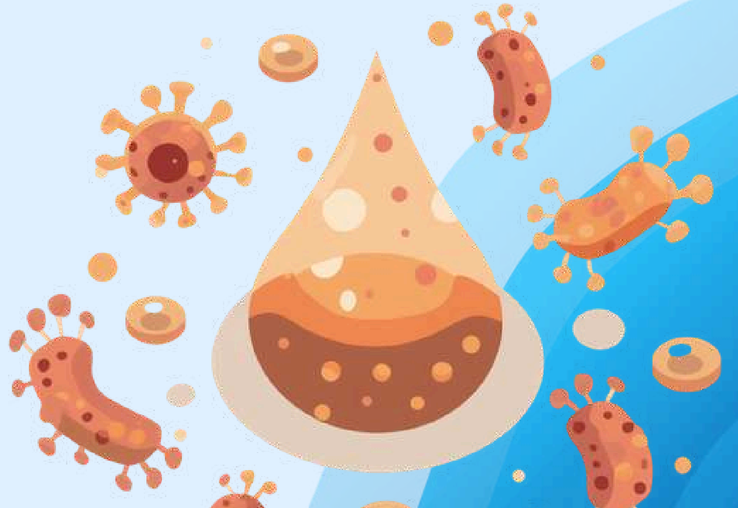
Water can contain harmful substances that make it unsafe. Chemicals like pesticides and industrial waste pollute our rivers. Microorganisms such as bacteria and viruses can cause serious illnesses. Heavy metals including lead and mercury are toxic to our bodies. Physical debris like plastic and sediments block water flow and harm aquatic life.



How Water Quality Affects Health

Clean Water Benefits: Drinking safe, clean water keeps our bodies healthy and strong. It helps prevent diseases like cholera, typhoid, and diarrhea. Clean water supports proper digestion, healthy skin, and strong immunity.

Contaminated Water Risks: Drinking polluted water can cause serious health problems. Bacteria and chemicals lead to stomach illnesses, vomiting, and diarrhea. Heavy metals can cause skin problems, rashes, and long-term damage to organs.



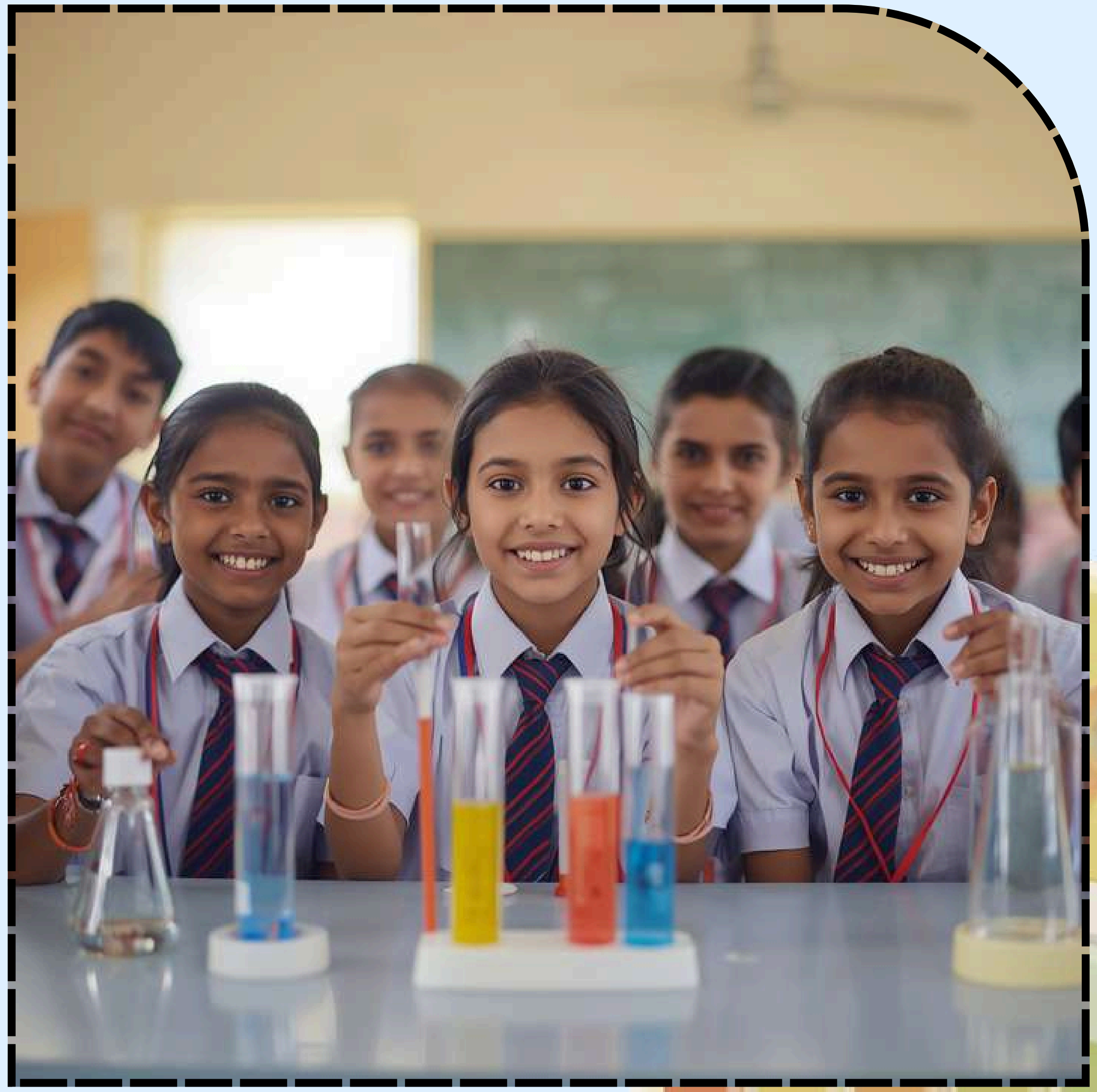
Basics of Water Testing

Testing water quality involves careful steps to ensure accurate results. First, collect samples safely using clean containers from different water sources. Scientists and students then examine the water for several key factors:

• Color and clarity - is it clear or cloudy?
• Smell - does it have any unusual odors?
• pH level - is it acidic, neutral, or alkaline?
• Chemical content - are there harmful substances?
• Microbes - are bacteria or viruses present?

Tools for Water Testing

Scientists and students use special tools to check if water is safe. pH strips change color to show if water is acidic or alkaline. A turbidity tube measures how clear the water is. Test kits can detect harmful bacteria we cannot see. Chemical test strips check for nitrates and chlorine levels.



Steps in Water Testing

Collect Sample: Gather water from your source using a clean container. Make sure not to touch the inside of the container or lid to avoid contamination.

Use Test Tools: Apply pH strips, turbidity tubes, or chemical test kits to your water sample. Follow the instructions carefully for each tool.

Record & Analyze: Write down all your measurements and observations. Compare results to safe water standards and decide if the water is safe to use.

Our Water Testing Experiment

Time to become real scientists! In this hands-on activity, we will collect water samples from different sources around our school and homes. Using our testing tools, we will check pH levels, clarity, and look for contaminants. Record all your observations carefully on your worksheets!

Reading Test Results

Understanding your water test results is easy! The pH scale runs from 0-14: below 7 is acidic (like lemon juice), 7 is neutral (safe drinking water), and above 7 is alkaline. Safe drinking water should have pH between 6.5-8.5.

Turbidity measures water clarity in NTU units. Clear water reads 0-1 NTU, while anything above 4 NTU may be unsafe. For chemicals, safe chlorine levels are 0.2-4 mg/L, nitrates should stay below 10 mg/L, and lead must be under 0.01 mg/L.

What To Do If Water Is Contaminated?

Inform Adults and Authorities: If you find contaminated water, immediately tell your parents, teachers, or local health officials. They can help alert the community and take action to protect everyone.

Use Water Filters or Boil Water: Before drinking or cooking, always filter water through a clean cloth or water purifier. Boiling water for at least 1 minute kills harmful germs and makes it safer to use.

Avoid Unsafe Water and Support Clean-Up: Never drink or cook with water you suspect is contaminated. Join local clean-up efforts to remove pollution from water sources and protect your community's health.

Local Water Testing Story

In Rajasthan's Alwar district, students from a local school partnered with community elders to test well water across their village. Using simple pH strips and turbidity tubes, they discovered that three wells had unsafe contamination levels. Their findings led to the installation of water filters, and within months, cases of waterborne illness dropped by 60%. This student-led initiative inspired neighboring villages to start their own water testing programs, proving that young scientists can make a real difference in community health.

Benefits of Clean Water

Clean water can prevent up to 80% of diseases in children, including diarrhea, cholera, and typhoid. Access to safe water means healthier communities and brighter futures.

When families have clean water, children can attend school regularly, parents can work without illness, and communities thrive together. Every drop of clean water matters!

Let's Review!

What is one common water contaminant?
Name a tool used to test water pH.

Why is clean water important?
Think about what you learned today!

Thank You & Keep Exploring!

Thank you! Keep testing and protecting our water.